

- The ratio of the area of a circle to the square of its radius is equal to π .

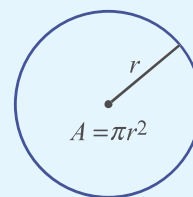
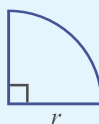
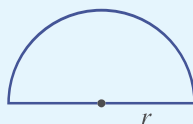
$$\text{i.e. } \frac{A}{r^2} = \pi \quad \text{so} \quad A = \pi r^2$$

- A half circle is called a **semicircle**.

$$A = \frac{1}{2} \pi r^2$$

- A quarter circle is called a **quadrant**.

$$A = \frac{1}{4} \pi r^2$$



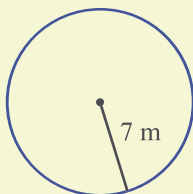
Key ideas

Example 10 Finding circle areas without technology

Find the area of these circles using the given approximate value of π .

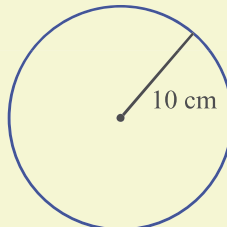
a

$$\pi = \frac{22}{7}$$



b

$$\pi = 3.14$$



SOLUTION

$$\begin{aligned} \text{a } A &= \pi r^2 \\ &= \frac{22}{7} \times 7^2 \\ &= 154 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{b } A &= \pi r^2 \\ &= 3.14 \times 10^2 \\ &= 314 \text{ cm}^2 \end{aligned}$$

EXPLANATION

Always write the rule.

Use $\pi = \frac{22}{7}$ and $r = 7$.

$$\frac{22}{7} \times 7 \times 7 = 22 \times 7$$

Use $\pi = 3.14$ and substitute $r = 10$.

3.14×10^2 is the same as 3.14×100

Example 11 Finding circle areas using a calculator

Use a calculator to find the area of this circle correct to two decimal places.



SOLUTION

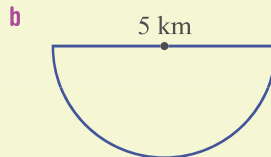
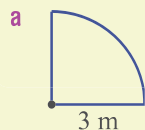
$$\begin{aligned} A &= \pi r^2 \\ &= \pi \times 2^2 \\ &= 12.57 \text{ cm}^2 \end{aligned}$$

EXPLANATION

Use the π button on the calculator and enter $\pi \times 2^2$ or $\pi \times 4$.

Example 12 Finding areas of semicircles and quadrants

Find the area of this quadrant and semicircle correct to two decimal places.

**SOLUTION**

$$\begin{aligned} \mathbf{a} \quad A &= \frac{1}{4} \times \pi r^2 \\ &= \frac{1}{4} \times \pi \times 3^2 \\ &= 7.07 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \mathbf{b} \quad r &= \frac{5}{2} = 2.5 \\ A &= \frac{1}{2} \times \pi r^2 \\ &= \frac{1}{2} \times \pi \times 2.5^2 \\ &= 9.82 \text{ km}^2 \end{aligned}$$

EXPLANATION

The area of a quadrant is $\frac{1}{4}$ the area of a circle with the same radius.

The radius is half the diameter.

The area of a semicircle is $\frac{1}{2}$ the area of a circle with the same radius.

Exercise 4E

- 1** Evaluate without the use of a calculator.

a 3.14×10

b 3.14×4

c $\frac{22}{7} \times 7$

d $\frac{22}{7} \times 7^2$

- 2** Use a calculator to evaluate these to two decimal places.

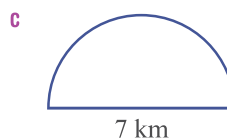
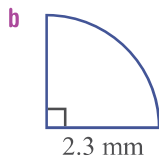
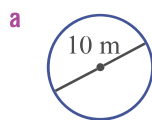
a $\pi \times 5^2$

b $\pi \times 13^2$

c $\pi \times 3.1^2$

d $\pi \times 9.8^2$

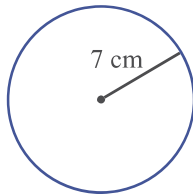
- 3** What is the length of the radius in these shapes?



Example 10

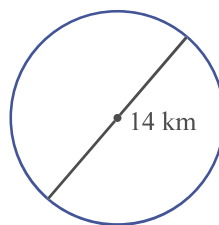
- 4 Find the area of these circles, using the given approximate value of π .

a



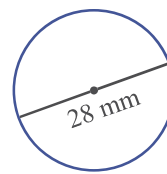
$$\pi = \frac{22}{7}$$

b



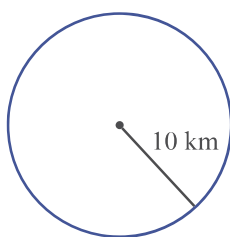
$$\pi = \frac{22}{7}$$

c



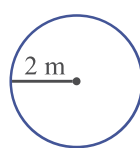
$$\pi = \frac{22}{7}$$

d



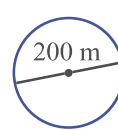
$$\pi = 3.14$$

e



$$\pi = 3.14$$

f



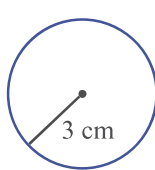
$$\pi = 3.14$$

Example 11

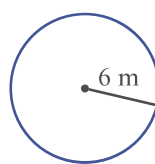
- 5 Use a calculator to find the area of these circles correct to two decimal places.



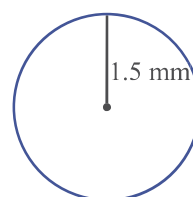
a



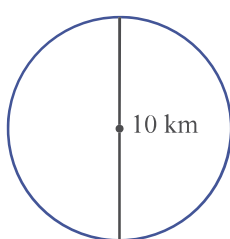
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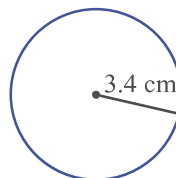
c



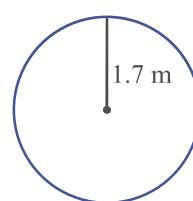
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**Example 12**

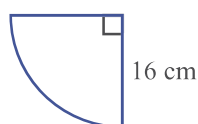
- 6 Find the area of these quadrants and semicircles correct to two decimal places.



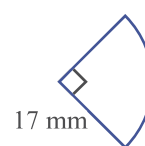
a



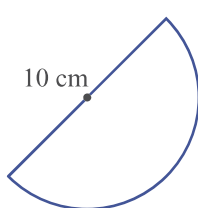
b



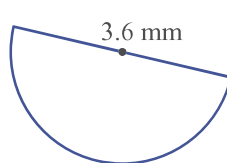
c



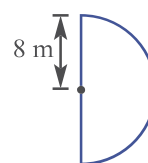
d



e



f





- 7 A pizza tray has a diameter of 30 cm. Calculate its area to the nearest whole number of cm^2 .



- 8 A tree trunk is cut to reveal a circular cross-section of radius 60 cm. Is the area of the cross-section more than 1 m^2 and, if so, by how much? Round your answer to the nearest whole number of cm^2 .



- 9 An circular oil slick has a diameter of 1 km. The newspaper reported an area of more than 1 km^2 . Is the newspaper correct?



- 10 Two circular plates have radii 12 cm and 13 cm. Find the difference in their area correct to two decimal places.



- 11 Which has the largest area, a circle of radius 5 m, a semicircle of radius 7 m or a quadrant of radius 9 m?



- 12 A square of side length 10 cm has a hole in the middle. The diameter of the hole is 5 cm. What is the area remaining? Round the answer to the nearest whole number.

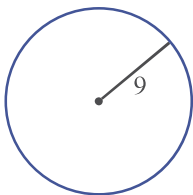


- 13 A circle has radius 2 cm.

- Find the area of the circle using $\pi = 3.14$.
- Find the area if the radius is doubled to 4 cm.
- What is the effect on the area if the radius is doubled?
- What is the effect on the area if the radius is tripled?
- What is the effect on the area if the radius is quadrupled?
- What is the effect on the area if the radius is multiplied by n ?

- 14 The area of a circle with radius 2 could be written exactly as $A = \pi \times 2^2 = 4\pi$. Write the exact area of these shapes.

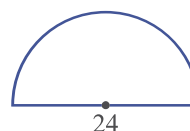
a



b



c



- 15 We know that the diameter d of a circle is twice the radius r , i.e. $d = 2r$ or $r = \frac{1}{2}d$.

- Substitute $r = \frac{1}{2}d$ into the rule $A = \pi r^2$ to find a rule for the area of a circle in terms of d .
- Use your rule from part a to check that the area of a circle with diameter 10 m is $25\pi \text{ m}^2$.



Enrichment: Reverse problems

- 16 Reverse the rule $A = \pi r^2$ to find the radius in these problems.

- If $A = 10$ use your calculator to show that $r \approx 1.78$.
- Find the radius of circles with these areas. Round the answer to two decimal places.
 - 17 m^2
 - 4.5 km^2
 - 320 mm^2
- Can you write a rule for r in terms of A ? Check that it works for the circles defined in part b.